



University of Alkafeel, College of Dentistry, Year 3, Course 1

Pathology Lab 5 Chronic Inflammation

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Learning Objectives

- ☐ **Know what Chronic Inflammation is.**
- ☐ **Know the causes of Chronic Inflammation.**
- ☐ **Know the Hallmarks of Chronic Inflammation.**
- ☐ **Know what the Special Pattern of Chronic Inflammation and its types.**

Introduction

Chronic inflammation is a state of **prolonged inflammation (weeks, months, or years)** in which **active inflammation, tissue destruction, and attempts at tissue repair** (angiogenesis and fibrosis) all occur simultaneously.

Causes of Chronic

Inflammation

1-Persistent Infections: By certain difficult-to-eradicate microorganisms, such as *Mycobacterium tuberculosis* (tuberculosis), *Treponema pallidum* (syphilis).

2-Prolonged Exposure to Potentially Toxic Agents:

- Exogenous:** Non-degradable inhaled materials like asbestos (causing asbestosis).

- Endogenous:** Excessive cholesterol and lipids in the arterial wall (contributing to atherosclerosis).

3-Autoimmunity/Hypersensitivity: Diseases where the immune system attacks the body's own tissues due to an inappropriate and sustained response. Rheumatoid Arthritis (RA), Systemic Lupus Erythematosus (SLE)

Hallmarks of Chronic Inflammation

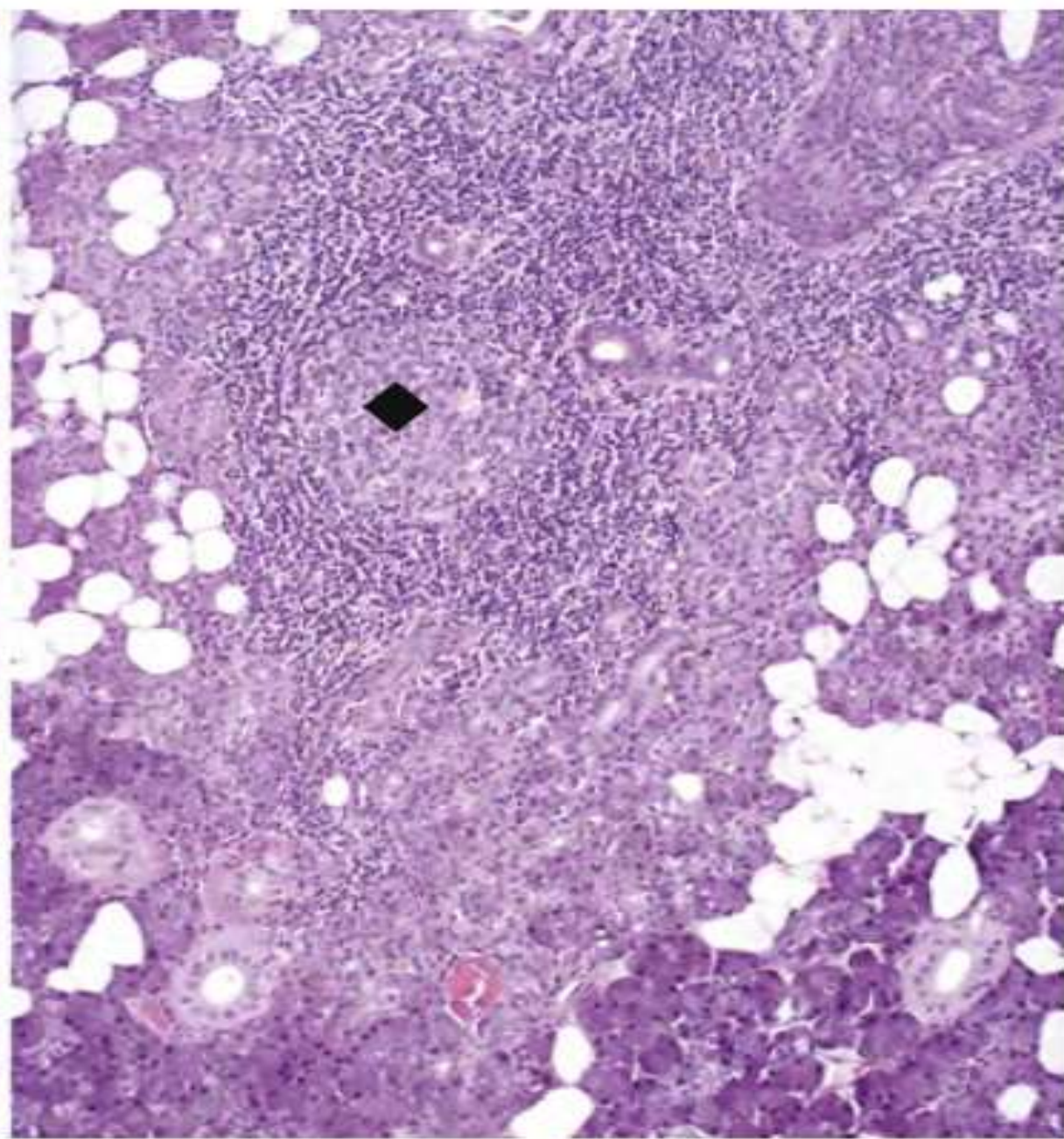
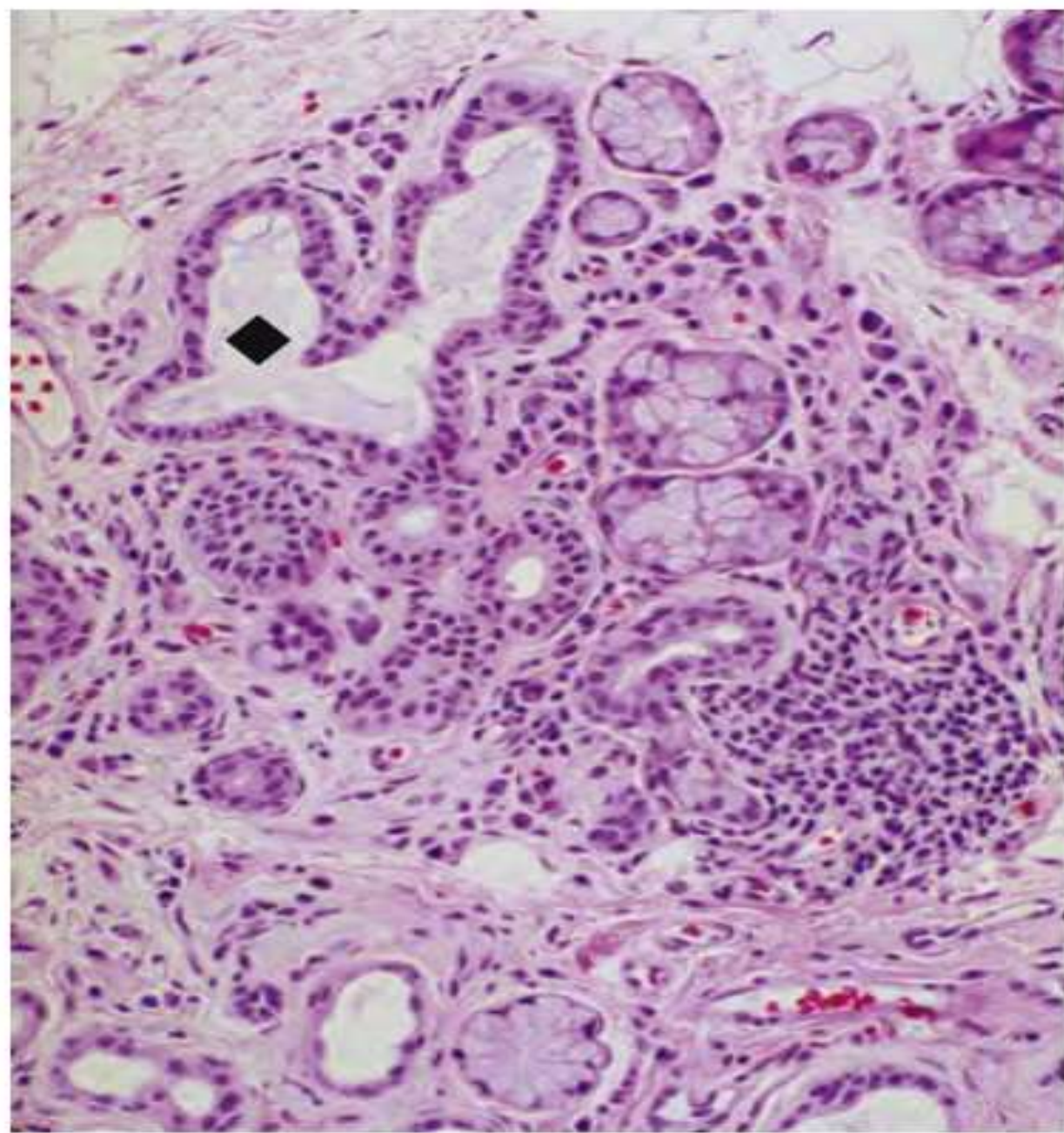
The cellular infiltrate in chronic inflammation is different from the neutrophil-rich exudate of acute inflammation.

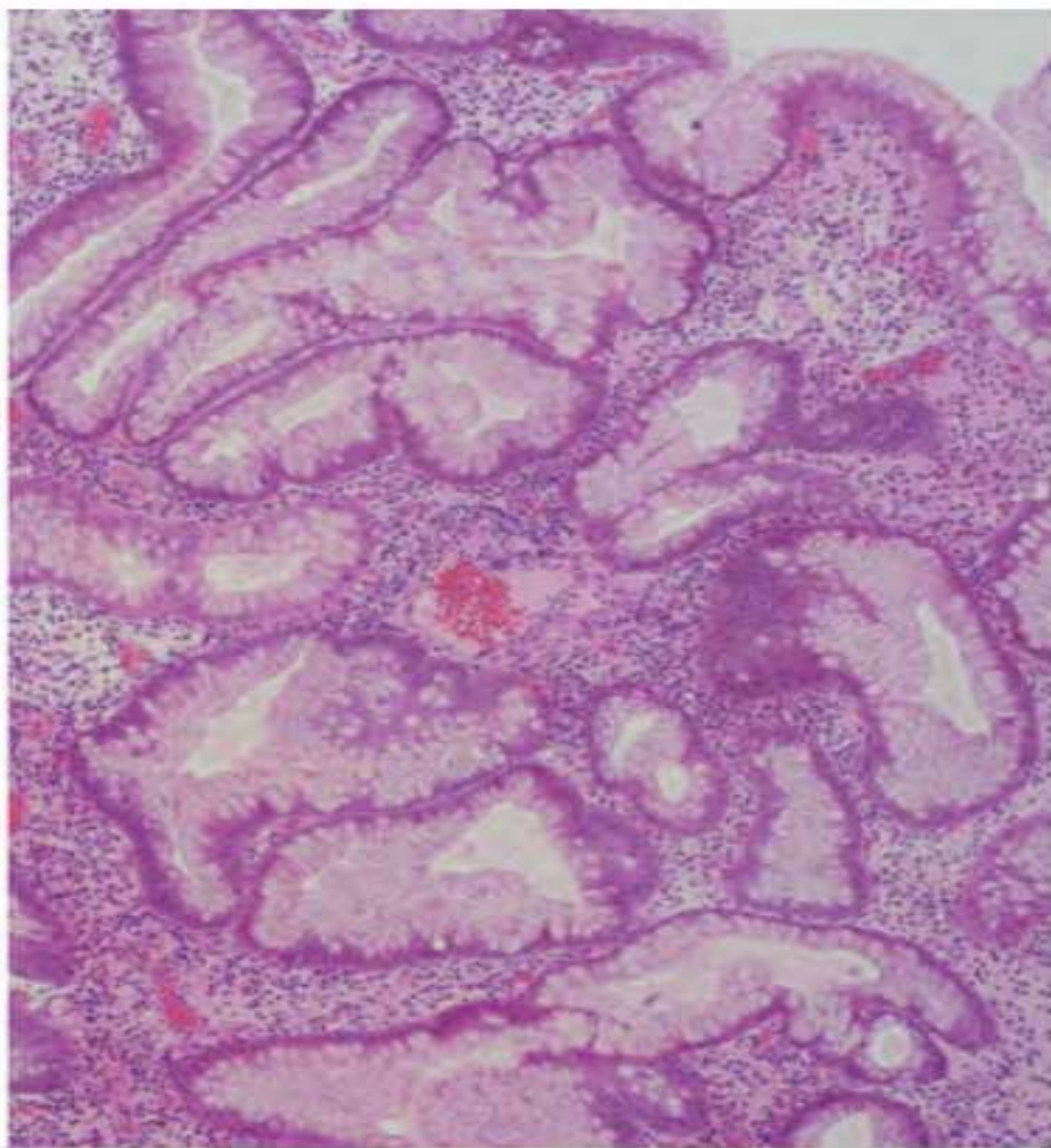
The defining microscopic features are:

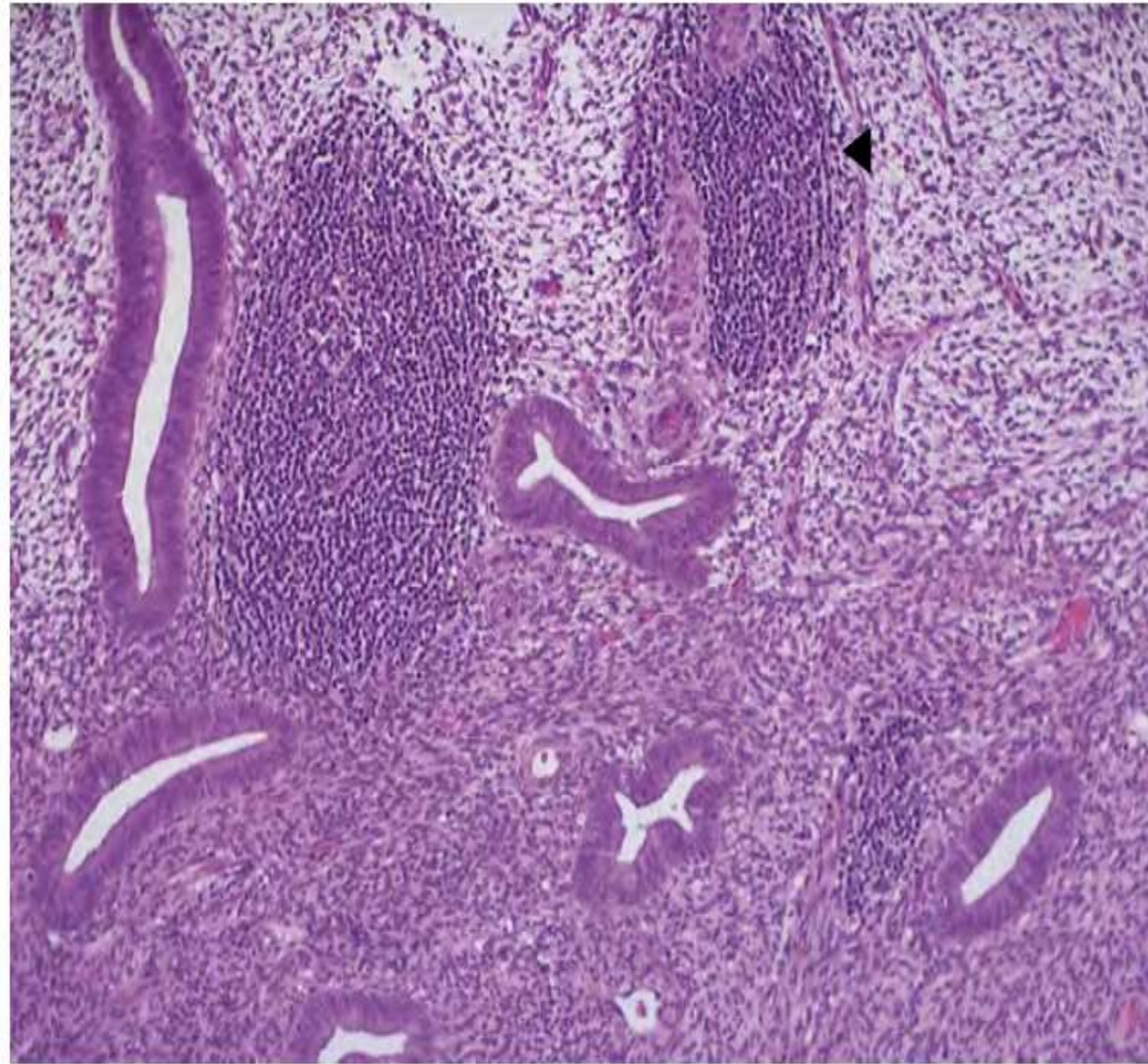
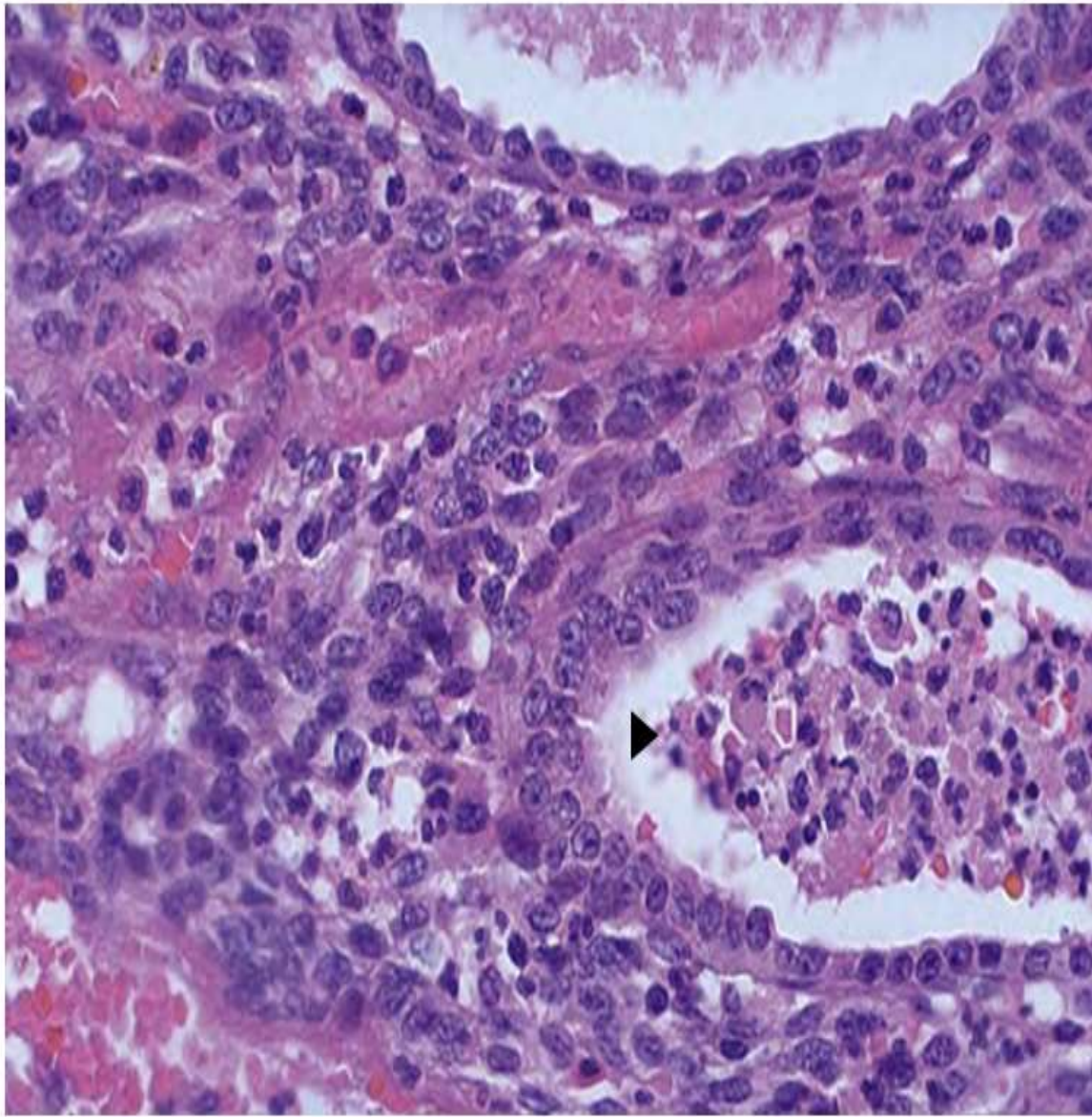
1-Infiltration with Mononuclear Cells: The primary cells are **macrophages, lymphocytes** (T and B cells), and **plasma cells**.

2-Tissue Destruction: Largely induced by toxic products (e.g. proteases) released by activated macrophages and other inflammatory cells.

3-Repair (Fibrosis and Angiogenesis):







Special Pattern

Granulomatous Inflammation

This is a distinctive, specialized pattern of chronic inflammation encountered when the inciting agent (e.g., certain bacteria, foreign body) is particularly difficult to eradicate by typical macrophage or acute inflammatory response.

Pathologic Features

A granuloma is a focal aggregate of **activated macrophages** which undergo a transformation to become **epithelioid cells**, surrounded by a collar of **lymphocytes** and **plasma cells**.

1-Epithelioid Macrophages: Macrophages that have elongated, slipper-shaped nuclei, vaguely resembling epithelial cells (hence the name).

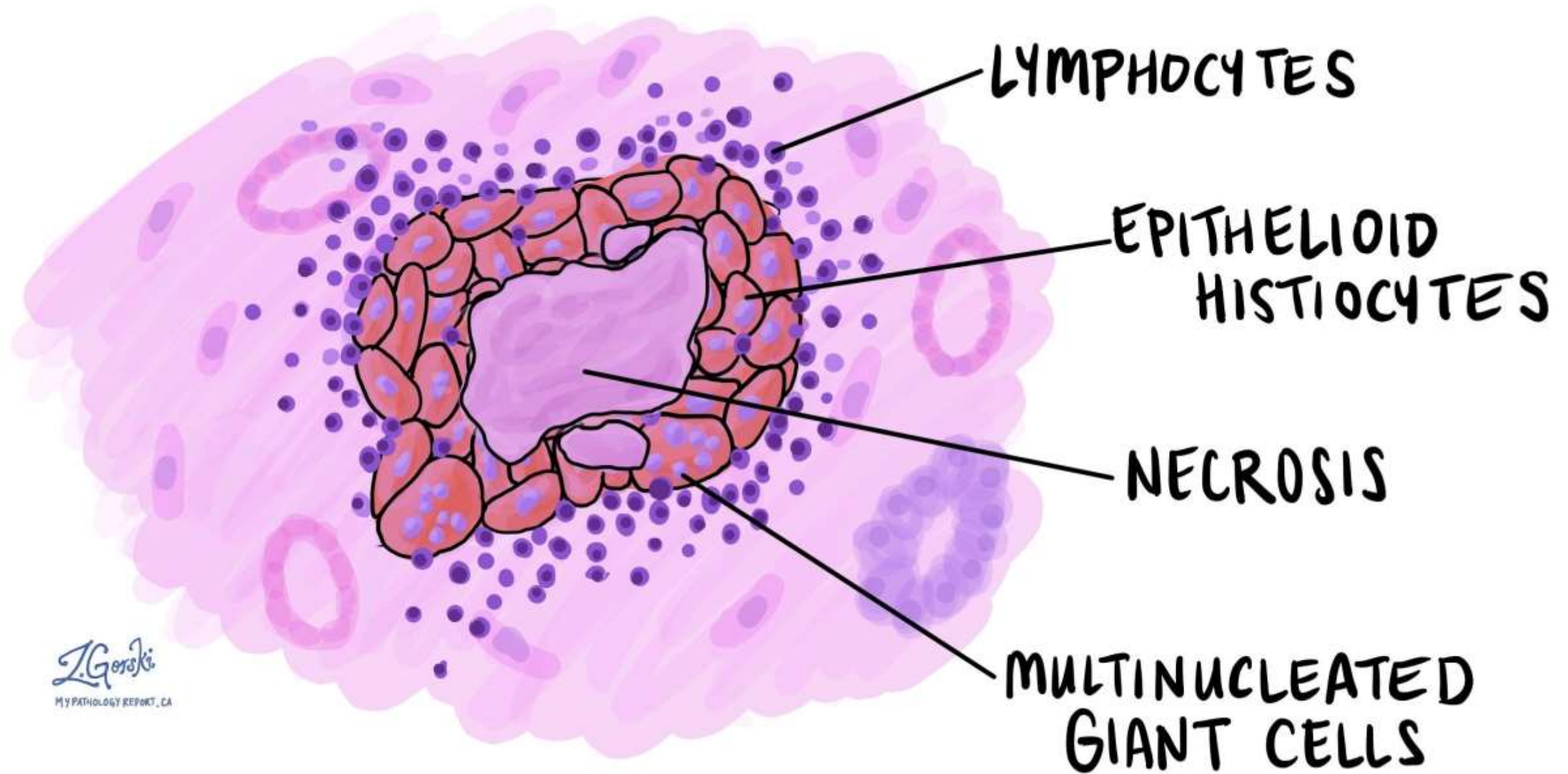
2-Giant Cells (Multinucleated Giant Cells): Formed by the fusion of multiple epithelioid macrophages.

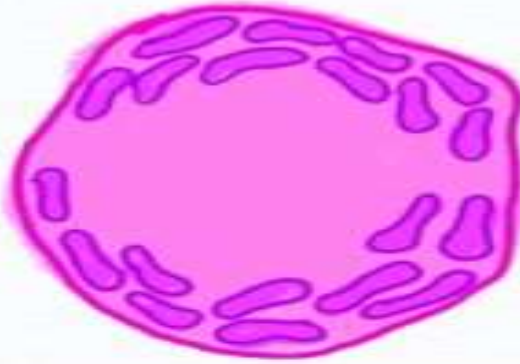
- Langhans Giant Cell:** Nuclei are typically arranged in a horseshoe or ring pattern near the periphery. Highly characteristic of tuberculosis.

- Foreign Body Giant Cell:** Nuclei are typically scattered randomly throughout the cell.

3-Lymphocyte Cuff: A rim of T-lymphocytes and other mononuclear cells surrounds the central collection of epithelioid cells.

4-Fibrosis: Older granulomas may show an outer layer of fibroblasts and connective tissue (fibrosis) as an attempt to "wall off" the persistent agent





Cells with nuclei are arranged in 'HORSESHOE' shaped pattern. Can be seen in one or both the poles

Langhans

NOT Langerhans!

GIANT CELL



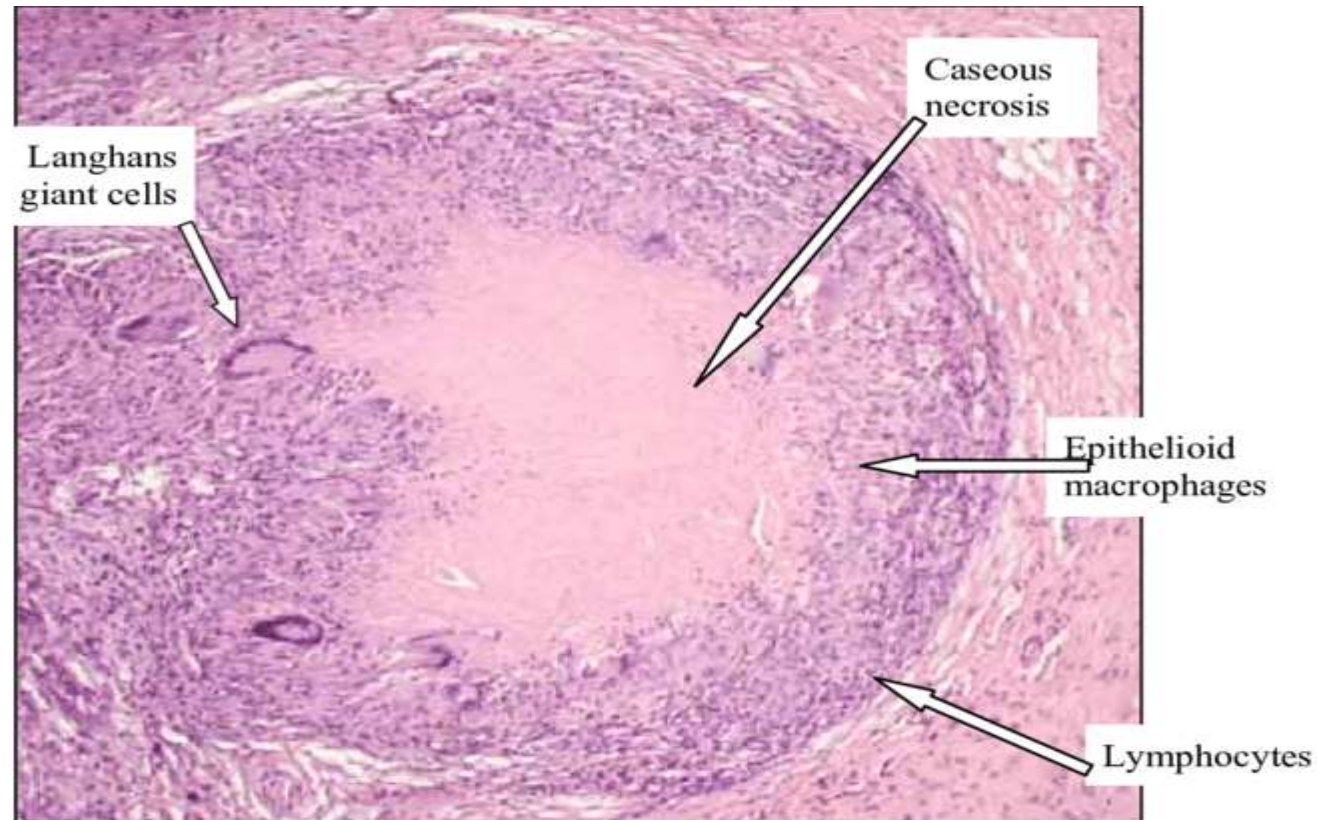
@Vijay

Contain regular nuclei scattered throughout cytoplasm

Foreign body

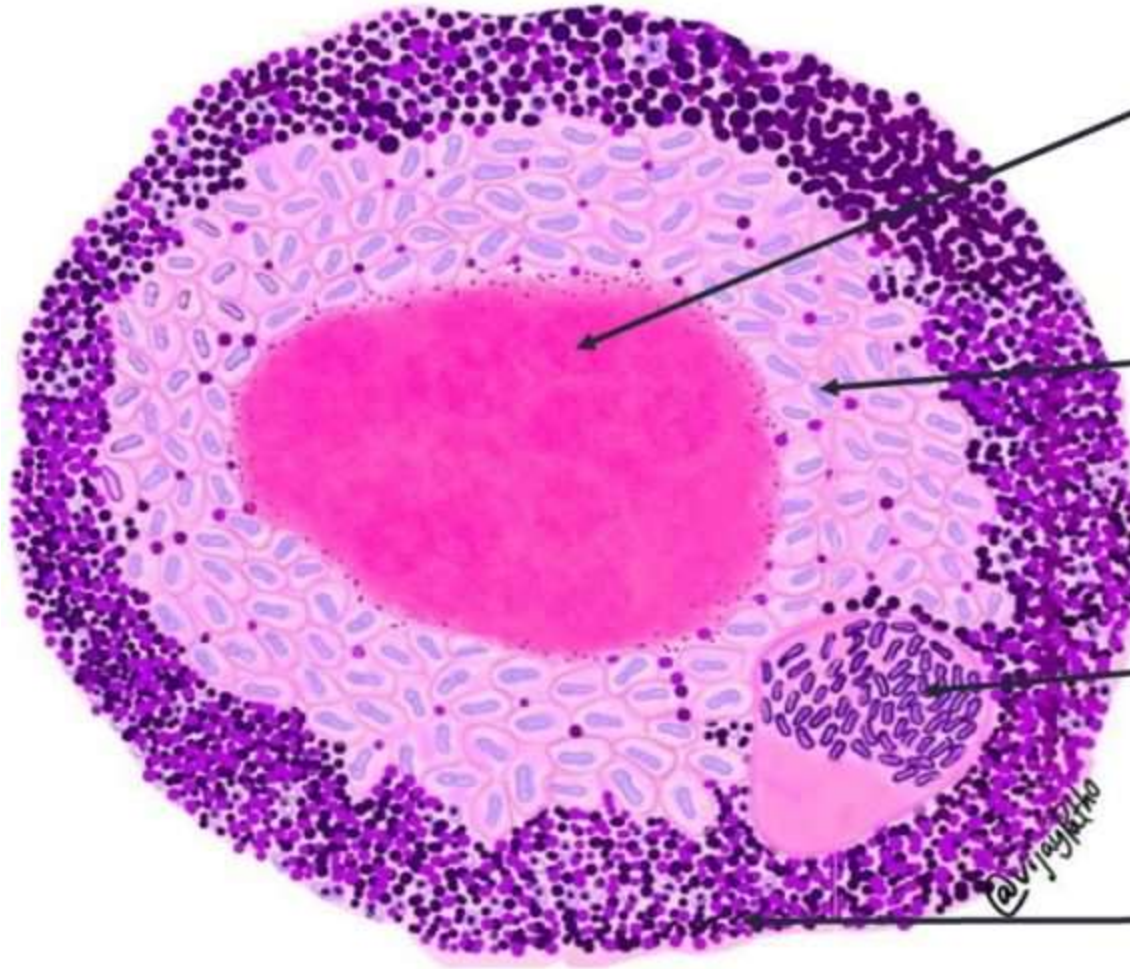
Classification of Granulomas

1-Caseating Granuloma: Shows a central zone of caseous necrosis (cheese-like amorphous debris, a hallmark of tuberculosis and some fungal infections).



TUBERCULOUS LYMPHADENITIS:

Necrotizing granuloma

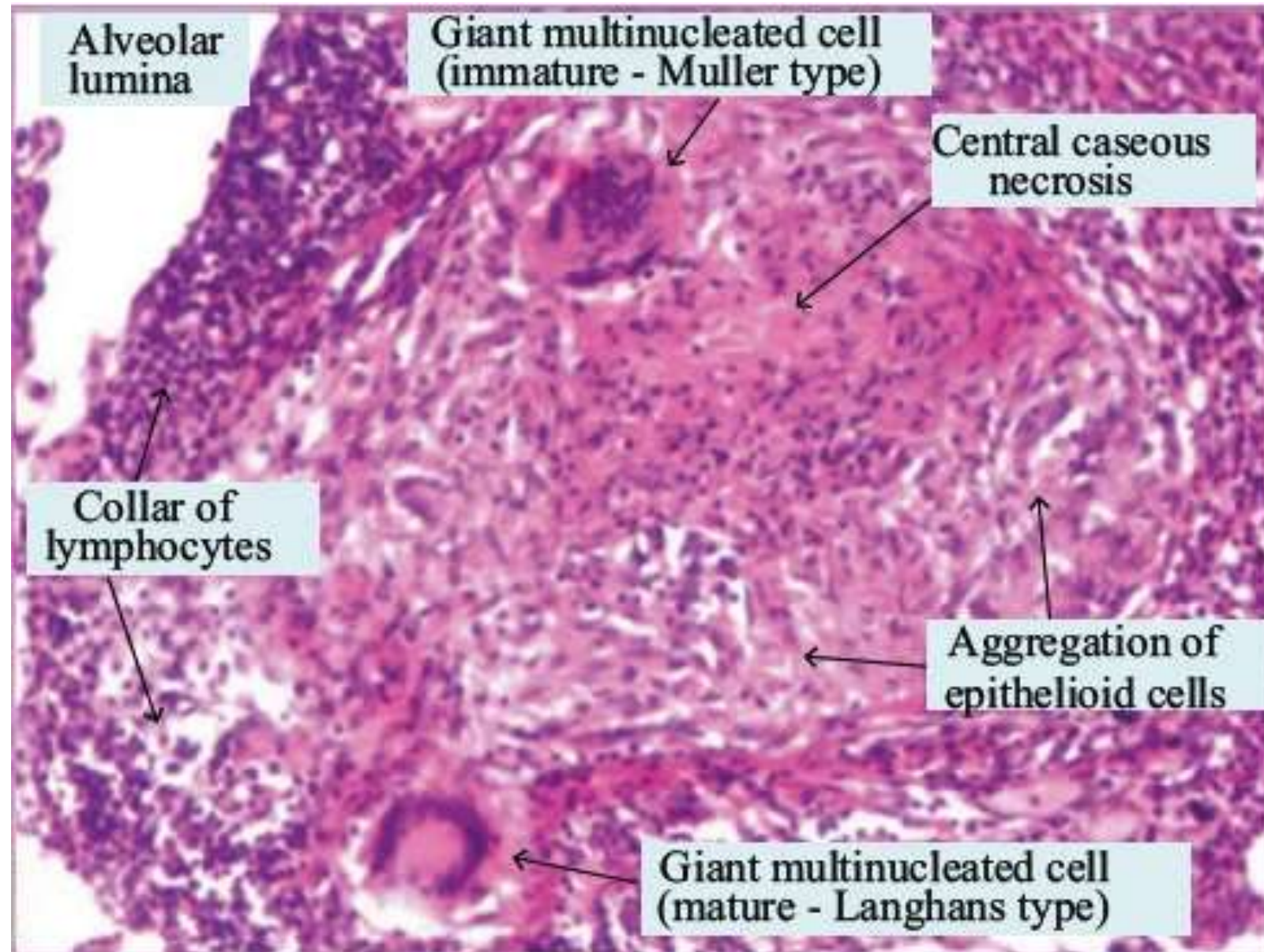


Amorphous granular
eosinophilic debritic material
CASEOUS NECROSIS

Modified macrophages with
abundant cytoplasm and pale
staining "slipper" shaped nuclei
EPITHELOID CELLS

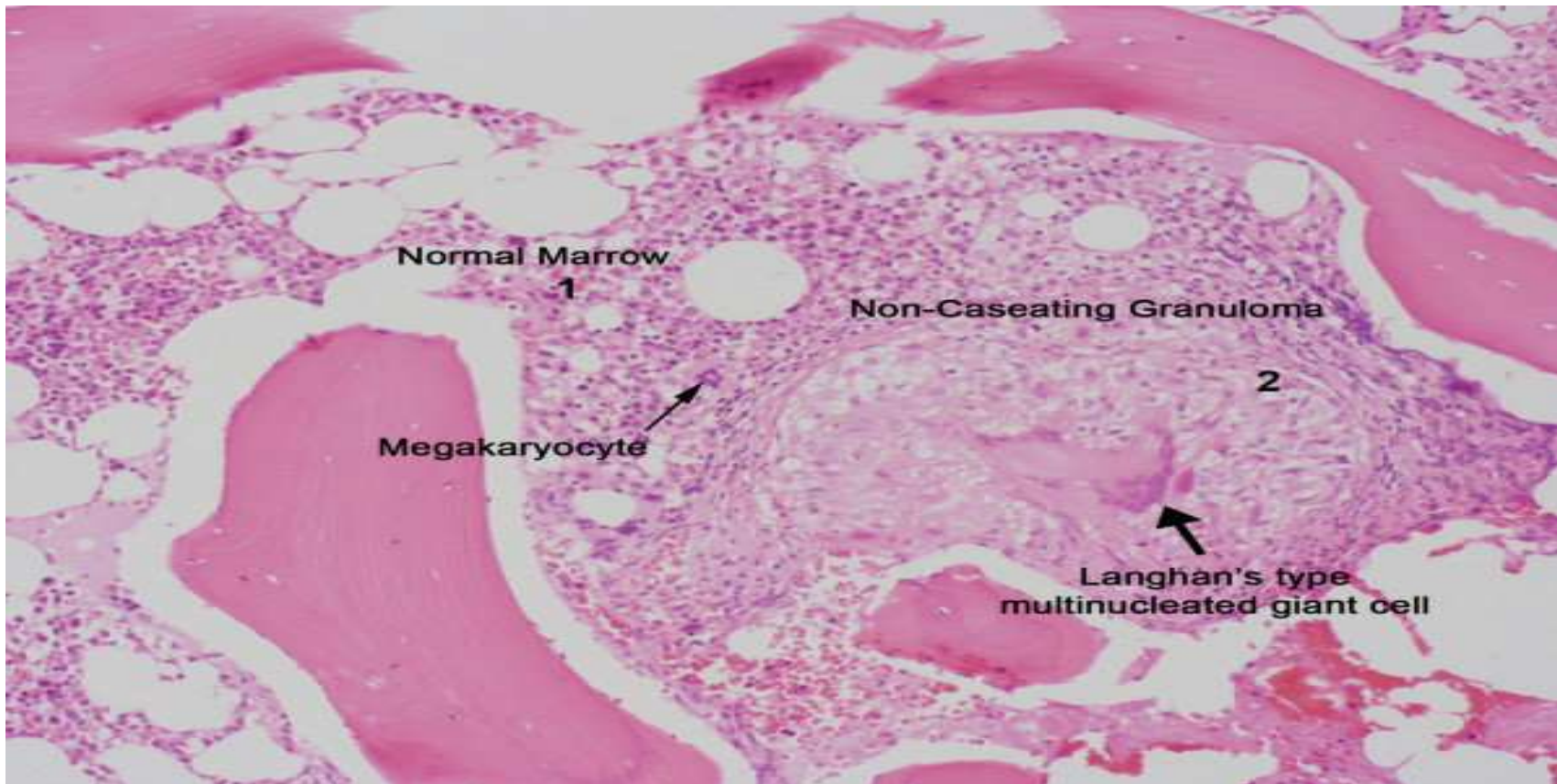
Multinucleated giant cell
LANGHAN GIANT CELL

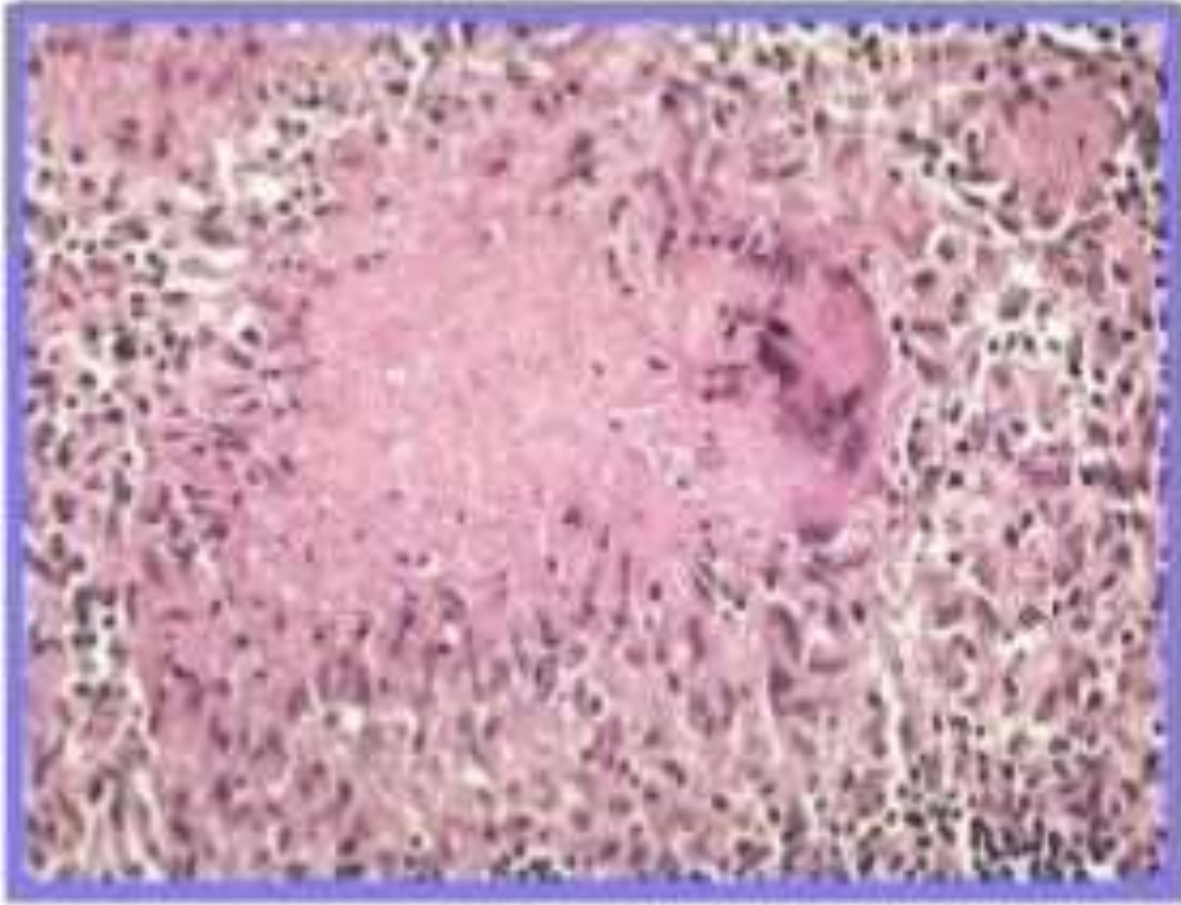
Collar of lymphocytes
surrounding epithelioid cell
aggregates



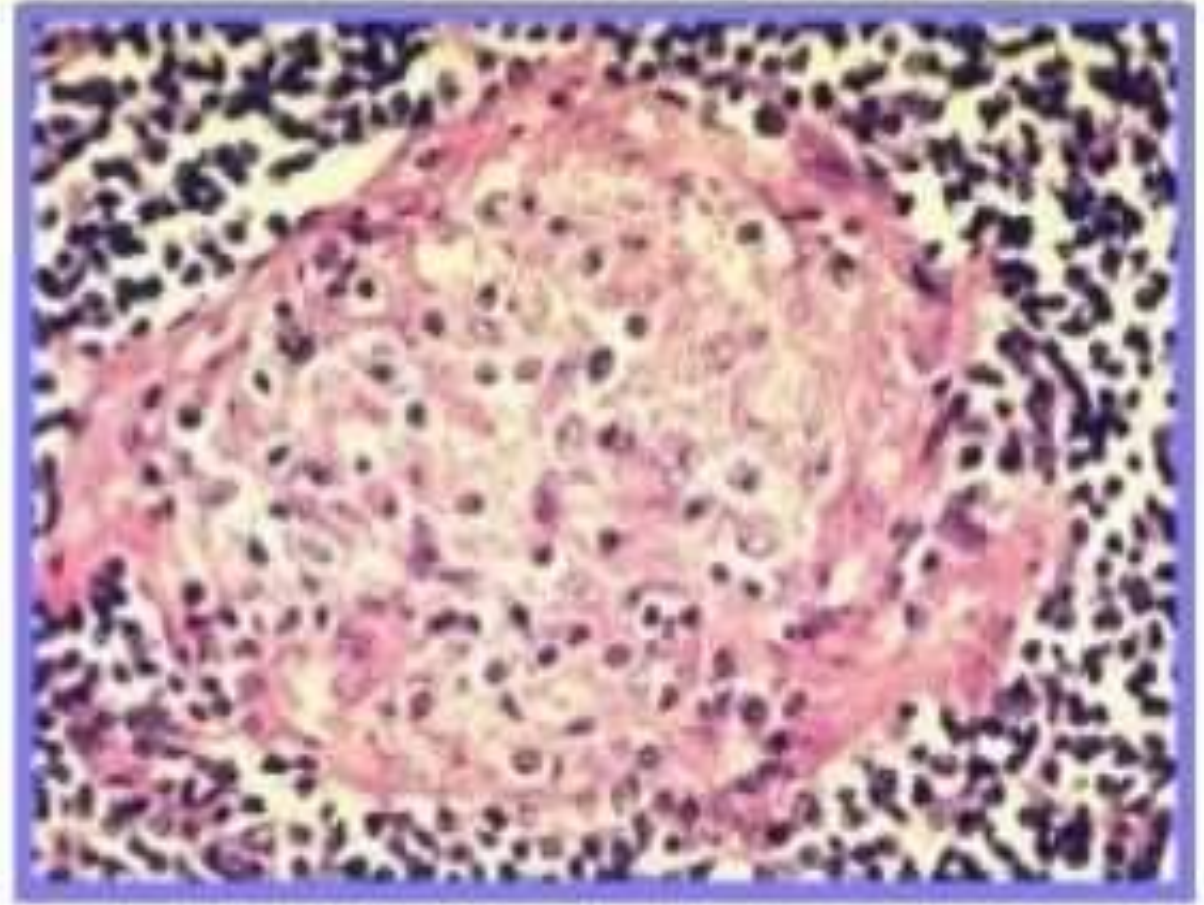
Classification of *Granulomas*

2-Non-Caseating Granuloma: Lacks the central necrosis (classic in sarcoidosis, Crohn's disease, and foreign body reactions).





Caseating granulomas
TB



Non caseating granulomas
Sarcoidosis

